

Polarization in SEDust: Implementation Notes

Abstract

These are maintenance notes for the polarization capability added to SEDust: where the polarized optics come from, how they are assembled, which implementation choices were made and why, and how far the numbers have been checked. The reader is assumed to be someone extending or repairing this code, not someone calling it. The calling interface is documented in `SEDust_user_manual.pdf`; the underlying physics is documented in `aligned_grain_polarization.pdf`. Neither is repeated here.

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1. Scope, and what lives elsewhere

Three documents cover this work and they do not overlap (Table 1). This one answers only the question *how was this built and why that way*. Where physics is needed to justify a coding choice, the relevant section of the theory report is cited rather than restated.

Question	Document
Why polarization at all; alignment theory; the polarized transfer equation; sign and index conventions; validation targets	aligned_grain_polarization.pdf (§1–§5, §9–§11)
How an RT code calls the library: signatures, units, what is already integrated	SEDust_user_manual.pdf, §5
How the code is built, why each choice was made, what was measured	this note

Table 1: Division of the polarization documentation.

Where this is going. The polarization capability is not an end in itself. The destination is the polarized radiative transfer of Seon (2018): dust scattering plus dichroic extinction by aligned grains, computed with grain optics derived from first principles, in the Monte Carlo formalism of Peest et al. (2017) as extended to aligned spheroids by Peest et al. (2023). Under that contract MoCafe supplies the alignment assumptions (the f_{align} profile and the local magnetic-field geometry) and SEDust returns every material quantity in matrix form: the 4×4 extinction matrix $K(\theta_i)$ with dichroism and birefringence, the phase matrix $Z_{\text{al}}(\theta_i; \theta_s, \phi)$ of the aligned population, the random-orientation remainder, and the

scattering cross sections. Dichroic extinction, polarized emission, birefringence, and the orientation-resolved cross-section table cover the extinction and emission side of that contract; the scattering side — the fixed-orientation Mueller matrix, the extinction matrix built from forward amplitudes, and their library interface — is now built and verified, and is documented in §8.. What remains is MoCafe’s consumption: the frame rotations, direction sampling, peel-off, and the $\exp(-K\tau)$ transfer step, none of which is a SEDust responsibility.

2. The data source

Everything polarized derives from one file, the orientation-resolved Draine & Hensley (2021) astrodust spheroid table:

```
data/dielectric/q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz
```

(porosity $P = 0.20$, iron fraction $f_{\text{Fe}} = 0$, axial ratio $b/a = 1.400$). Its layout is not self-describing, so it is recorded here (Table 2).

Property	Value
header	12 lines, skipped
payload order	((Q(jw,jr,jori), jw=0,1128), jr=0,168), jori=1,3
repeated	three times: Q_{ext} , then Q_{abs} , then Q_{sca}
one record on disk	the 169 sizes of one $(jw, jori)$ pair
records read	$3 \text{ quantities} \times 3 \text{ orientations} \times 1129$
total values	$3 \times 3 \times 1129 \times 169 = 1,717,209$
wavelength axis	data/dielectric/DH21_wave, 1129 nodes
size axis	data/dielectric/DH21_aeff, 169 nodes
axis file format	two title lines, then the values free-format

Table 2: Layout of the orientation-resolved Q table. The wavelength and size axes are *not* parsed out of the header; they come from the companion grid files, which list the nodes the table was computed on.

Orientation index. The convention is the trap most likely to bite a future reader, and it is worth stating in full (Table 3); see also §3.6 of the theory report.

jori	geometry	meaning
1	$\mathbf{k} \parallel \hat{a}$	propagation along the symmetry axis
2	$\mathbf{k} \perp \hat{a}, \mathbf{E} \parallel \hat{a}$	field along the symmetry axis
3	$\mathbf{k} \perp \hat{a}, \mathbf{E} \perp \hat{a}$	field across the symmetry axis

Table 3: Orientation index of the DH21 table. $\hat{a}$ is the spheroid symmetry axis.

3. Derived quantities

For a grain whose symmetry axis lies in the plane of the sky and which is perfectly aligned, the polarization cross section is the difference the grain presents to the two linear polarizations, halved:

$$Q_{\text{pol}} = \frac{1}{2} [Q(\text{jori}=3) - Q(\text{jori}=2)], \quad (1)$$

formed separately from Q_{ext} and from Q_{abs} , giving $Q_{\text{pol,ext}}$ and $Q_{\text{pol,abs}}$. The random-orientation average of the same table is the trace,

$$Q_{\text{ran}} = \frac{1}{3} [Q_1 + Q_2 + Q_3], \quad (2)$$

which is the modified picket fence approximation (MPFA; §3.2 of the theory report). Cross sections follow from

$$C = Q \pi a_{\text{eff}}^2, \quad (3)$$

with a_{eff} converted to cm. The alignment efficiency is the Hensley & Draine (2023) Table 1 fit,

$$f_{\text{align}}(a) = \frac{f_{\text{max}}}{1 + (a_{\text{align}}/a)^\alpha}, \quad a_{\text{align}} = 0.0749 \mu\text{m}, \quad \alpha = 1.80, \quad f_{\text{max}} = 1.00. \quad (4)$$

4. Module map

Table 4 lists what each file is responsible for. Nothing else in the tree touches polarization.

5. Implementation decisions

This is the part worth reading again in six months.

5.1 The accumulation sites, and the two that hide

`sed_grain_loop` branches on `stoch_method` into four cases, and the polarized channel has to be accumulated wherever the total is. There are eight such statements (Table 5). Missing any one of them produces a polarized spectrum that is silently too low rather than an error.

The lesson. The `'draine'` and `'qm'` branches are OpenMP-parallel and accumulate into a thread-local `Jout_local` that is reduced at the end. The `'heuristic'` branch is serial and accumulates into `Jout` directly. A survey of the accumulation sites conducted by searching for `Jout_local` therefore returns only the parallel branches, and an initial pass over this code identified four sites on exactly that basis. Since `'heuristic'` is the *default* solver, that omission would have made the polarized emission come back identically zero in the default configuration while every other solver looked correct. Any future change that adds a term to the emission sum must be checked against Table 5 branch by branch, not by pattern search.

5.2 Why the 'qm' rescaling is exact

qm_solve_grain does not return a temperature distribution; it returns the emitted spectrum with C_{abs} already folded in. The polarized counterpart is therefore obtained by rescaling:

```
if (Cabs_pop(ii, ir) > 0.0_wp) then
  Jpol_local(ii) = Jpol_local(ii) + &
    wpol * emission_qm(ii) * (Cpol_pop(ii, ir) / Cabs_pop(ii, ir)) / &
    (4.0_wp * PI * lam(ii) * 1.0e-3_wp)
end if
```

This is not an approximation. The temperature distribution $P(T)$ is set by the *heating*, which is governed by C_{abs} and is unchanged by the question of which polarization is being read out. Only the emission readout changes, so

$$C_{\text{pol}} \int P(T) B_{\lambda}(T) dT = \frac{C_{\text{pol}}}{C_{\text{abs}}} C_{\text{abs}} \int P(T) B_{\lambda}(T) dT = \frac{C_{\text{pol}}}{C_{\text{abs}}} \text{emission_qm} \quad (5)$$

holds identically, wavelength by wavelength. Wavelengths with zero C_{abs} emit nothing and are skipped rather than divided by zero.

5.3 The $P(T)$ solver was not touched

No change was made to calc_P, narrow_iterative, or qm_solve_grain. This is deliberate and it is what makes stochastic heating automatically correct for polarization: at every site the same temperature weight multiplies both accumulators, one with $dn C_{\text{abs}}$ and the other with $dn f_{\text{align}} C_{\text{pol}}$. The polarized spectrum is thus the same average over the same $P(T)$, and no separate stochastic treatment of the polarized channel is needed (see §5.7 of the theory report for why this is the physically right structure).

5.4 Mixed orientation-average conventions

This is the one open physical wart, and it is documented rather than hidden.

C_{abs} in SEDust comes from our own T-matrix run, whose random-orientation average is computed exactly. C_{pol} comes from the release table, whose available average is the 1/3 trace, Eq. (2), i.e. the MPFA. The two are different approximations to the same quantity, and the code mixes them: build_Cpol adds the polarized channel and deliberately does *not* recompute C_{abs} .

Table 6 gives the measured disagreement, $|Q_{\text{abs}}^{\text{trace}}/Q_{\text{abs}}^{\text{exact}} - 1|$, on the shared $(\lambda, a_{\text{eff}})$ grid.

Reading the far-infrared maximum. The 3.46% worst case occurs at $a = 4.2 \mu\text{m}$, a size carrying $dn/n_{\text{H}} \sim 2.8 \times 10^{-36}$, so it is numerically irrelevant. Restricting to $a \leq 0.5 \mu\text{m}$, which covers 99.999% of the geometric cross section, the worst case drops to 1.98%. At $a \ll \lambda$ both averages approach the same Rayleigh limit, which is why the far infrared is well behaved.

Verdict: deliberately deferred in this release. The mixture is left in place for v1.20, as a decision rather than an oversight. The reasoning is recorded here so that it is not rediscovered later. The mixture is harmless for polarized

emission, which is a far-infrared phenomenon. It is **not** defensible for polarized *extinction* in the ultraviolet or the optical, and column 8 spans the whole wavelength grid. The `Cpol_ext` output of `dust_extinction` is the same quantity by the same route and carries the same caveat unchanged. Anyone using either shortward of the near infrared must resolve this first. The clean resolution is to recompute C_{abs} on the MPFA convention so that both channels share one average. That changes the existing SEDs, so it breaks their byte-identity with the current reference outputs and has to be done as a separate, deliberately validated step rather than folded into this one. The size of the change is not an argument for leaving it: the correctness argument is that two channels of one calculation should not use two different orientation averages.

What was decided for this release is only *when*, not *whether*. The planned use of these optics — redoing Seon (2018) in the *V* band, with one or two further optical bands possible — lies entirely in the range where the disagreement is 0.080% or below, while unifying the convention would change C_{abs} and break the byte-identical regression baseline that the rest of v1.20 was validated against. Spending that baseline buys nothing inside the present scope. It has to be spent before any ultraviolet polarized extinction is quoted, and that is the trigger for doing the work.

5.5 f_{align} is the fit, not the release column

`data/release/size_distribution.dat` also ships an alignment column. It agrees with Eq. (4) to a median ratio of 1.0000 but differs by up to 0.32% at the small end. The analytic form is the canonical one in SEDust because it is the published fit, because it is defined on any radius grid rather than only the tabulated one, and because it keeps the alignment model in a single place. Do not mix the two sources: a quantity computed half from one and half from the other is neither.

5.6 Names and surface area

Cpol was not renamed to Cpol_abs. The name appears at roughly 40 sites and is part of the public surface (`grain_pop_t%Cpol`, the optional `set_pop(..., Cpol_in=...)`). A rename would touch every one of them for no functional gain. The declaration carries an explicit note instead: `Cpol` is the absorption one, `Cpol_ext` its extinction twin.

The polarized extinction was exported as a table column first. A dedicated `dust_cpol_ext` accessor was considered at the time and rejected: `dust_lib` was then an emission-facing API with no extinction surface at all, so such an accessor would have been the module's only extinction entry point, a worse inconsistency than the inconvenience it removed. The condition attached to that decision was that `Cpol_ext` should join a full extinction API if one ever appeared. It has; see §5.7.

5.7 Why the accessor exists now

The table-only export was right for a library that emitted and did not extinguish. Rebuilding a radiative transfer code on top of SEDust changed the premise: the host would have obtained its emission from a call and its opacity from a

parsed text file, and that asymmetry is the whole problem. The file is also written on whatever wavelength grid the driver used, so a host on any other grid has to interpolate optics that the model could have evaluated exactly.

dust_extinction closes it. The model object already held everything required, so the accessor adds no physics and no new data source; it performs the same size integral the driver performs, over the populations already built. Two fields were added to grain_pop_t to make that possible: Cpol_ext, previously a module array only, and gsca, the scattering asymmetry, which the emission path never needed and which sed_init now interpolates onto the size grid beside C_{abs} and C_{sca} , from the same Q table and by the same routine the driver used. Agreement with the table is at the precision the file is written with (a median relative deviation of $\sim 7 \times 10^{-9}$ in each cross section), which is the expected result for one integral evaluated twice.

The table remains, unchanged byte for byte, and remains the right route for a tool that wants the optics without linking the library.

5.8 Compression handling, and a scratch file that leaked

The table ships compressed and Fortran cannot read a deflate stream. Adding a zlib binding was rejected as too heavy for one file. The reader instead shells out once,

```
call execute_command_line('gzip -dc "'//trim(gz_file)//'" > "'// &
                           trim(out_file)//'", exitstat=estat, cmdstat=cstat)
```

writes q_jori_scratch.dat in the working directory, reads it, and deletes it. A fresh clone works with no manual setup step, and neither a 17 MB untracked sibling in data/dielectric nor a new build dependency is created. A path not ending in .gz is opened directly.

The bug this had. Shell redirection creates the target file *before* gzip runs, so when decompression failed the early return left a zero-length q_jori_scratch.dat behind in the user's working directory. The failure path now calls discard_scratch_copy before bailing out, like every other error exit. The general shape of the bug, an error path that skips the removal the success path performs, is worth watching for in any future edit to this reader: it has eight separate early returns.

6. Verification

6.1 Polarized extinction against the release

The comparison against the release is restricted to $\lambda \geq 0.11 \mu\text{m}$ because the reference file turns negative below that. The 0.9422% maximum deviation is not a systematic offset: it sits at $\lambda = 0.112 \mu\text{m}$, right at the sign reversal of the polarized extinction, where the quantity passes through zero and a relative deviation is not a meaningful measure. Away from that point the agreement is at the level of the median.

The 7×10^{-8} figure is the precision the ASCII output format carries, so column 8 and calc_pol_ext.x agree as closely as the files allow. They are two independent routes to the same quantity: calc_pol_ext.x works straight from qpol_ext

and the release size distribution, while column 8 goes through `sed_init`, the $\log a$ interpolation onto the SED size grid, and `build_Cpol`.

6.2 Polarized emission

Intrinsic polarization fraction `lamI_pol/lamI_total` under a Mathis ISRF at $U = 1.585$:

The mid-infrared zero is correct rather than a failure: there the emission is dominated by stochastically heated PAHs, which HD23 take to be unaligned.

19.15% versus Planck’s 22% is a property of the model, not a bug. Planck measures $p_{\max} = 22.0(+3.5, -1.4)\%$ at 353 GHz. The 19.15% here is the *maximum* attainable value, obtained before any geometric factor, and every real line of sight reduces it, so the model as implemented cannot reach the observed maximum. This traces to the dust model itself: the axial ratio $b/a = 1.4$ is the minimum non-sphericity DH21b found the starlight polarization efficiency to require, and it was not tuned to maximize the submillimeter polarization fraction. Do not close the gap by adjusting f_{align} or the axial ratio without recognizing that as a change to the dust model, requiring its own revalidation against the extinction and emission constraints HD23 fit.

7. First-principles orientation-resolved optics

The table of §2. is Draine & Hensley’s precomputed one. SEDust can now regenerate the same orientation-resolved cross sections from the astro dust dielectric function with its own T-matrix engine, so the polarized optics no longer have to be taken on trust from the release file. The random-orientation optics were already SEDust’s own (the `q_astro dust` table read by `q_table`); this closes the remaining gap, the orientation-resolved part read by `q_table_jori`.

Recall the three orientations of the incident wave relative to the spheroid symmetry axis $\hat{a}$: `jori = 1` is $\mathbf{k} \parallel \hat{a}$; `jori = 2` is $\mathbf{k} \perp \hat{a}$ with $\mathbf{E} \parallel \hat{a}$; `jori = 3` is $\mathbf{k} \perp \hat{a}$ with $\mathbf{E} \perp \hat{a}$. The polarized and mean combinations are $Q_{\text{pol}} = \frac{1}{2}(Q_3 - Q_2)$ and $Q_{\text{ran}} = \frac{1}{3}(Q_1 + Q_2 + Q_3)$.

The optics are reached two ways. The *table* is loaded once and interpolated: this is the production path, fast and safe to call from many threads. The *direct* routine `oriented_cross_sections` computes one node from first principles; it is the shared core the table generator loops over, and it also serves an arbitrary (a, λ) or a shape or porosity for which no table exists. It must not be called from inside a loop over radiative-transfer cells: each call is a full T-matrix solve plus an angular integral, and the Mishchenko routines carry their state in COMMON blocks and are not thread-safe.

7.1 Rayleigh regime, $x < 0.1$

In the electrostatic limit the oblate spheroid has a diagonal polarizability $\alpha = \text{diag}(\alpha_a, \alpha_b, \alpha_b)$, with α_a for $\mathbf{E}$ along the symmetry axis and α_b for $\mathbf{E}$ transverse, set by the depolarization factors L_a, L_b of the spheroid through $\alpha_i \propto (\varepsilon - 1)/[1 + (\varepsilon - 1)L_i]$. Absorption and scattering follow as $Q_{\text{abs}} \propto \text{Im } \alpha$ and $Q_{\text{sca}} \propto |\alpha|^2$. The three orientations sample exactly these two polarizabilities: `jori = 1` and `3` present the transverse α_b (the field is transverse to $\hat{a}$ in both), `jori = 2` presents α_a . So the orientation-resolved cross sections are the two polarization components the random average $(\alpha_a + 2\alpha_b)/3$ is already

built from; extracting them before the average is analytic and exact, and $Q(\text{jori} = 1)$ equals $Q(\text{jori} = 3)$ identically. `rayleigh_limit` returns them through optional arguments, leaving its averaged outputs untouched.

7.2 T-matrix regime, $0.1 \leq x \leq 50$

The T-matrix $\mathbf{T}$ is a property of the particle alone — size, shape, refractive index — independent of orientation and of the incidence and scattering directions. SEDust solves it once for each wavelength and radius with Mishchenko's `tmd_one_scattermat`. Two orientation-resolved quantities are then read off the same $\mathbf{T}$.

Extinction, from the optical theorem. Mishchenko's AMPL evaluates the fixed-orientation amplitude matrix $\mathbf{S}(\mathbf{k} \rightarrow \mathbf{k}')$ from the stored $\mathbf{T}$ for a particle placed at chosen Euler angles. Taking the scattering direction equal to the incidence direction (forward) and applying the optical theorem, $C_{\text{ext}} = (4\pi/k) \text{Im} S_{\text{fwd}}$ for the incident polarization, gives C_{ext} for each orientation: the symmetry axis is placed along $\mathbf{k}$ ($\text{jori} = 1$) or across it ($\text{jori} = 2, 3$) and the co-polar forward amplitude is read ($S_{\theta\theta}$ for $\mathbf{E}$ along $\hat{\theta}$, $S_{\phi\phi}$ for $\hat{\phi}$).

Scattering, from the phase-matrix integral. The same AMPL, swept over all scattering directions at fixed incidence, gives the differential scattering cross section $|\mathbf{S}|^2$; the scattering cross section is its integral over the sphere, $C_{\text{sca}} = \int (|S_{\theta\leftarrow p}|^2 + |S_{\phi\leftarrow p}|^2) d\Omega$ for incident polarization p , and absorption follows by conservation, $C_{\text{abs}} = C_{\text{ext}} - C_{\text{sca}}$. The fixed-orientation scattered intensity is a trigonometric polynomial of degree $\sim 2N_{\text{max}}$ in $\cos\theta$ and in ϕ , so Gauss-Legendre nodes in $\cos\theta$ ($N_{\text{max}} + 2$ of them) and a uniform azimuth grid ($2N_{\text{max}} + 2$ points) integrate it to machine precision.

An engine subtlety. Mishchenko's random-orientation expansion GSP reuses the `/TMAT/` storage as scratch through an EQUIVALENCE and so destroys the T-matrix it reads. The amplitude evaluation needs that same T-matrix afterward, so `tmd_one_scattermat` keeps an intact copy in a second common block and restores `/TMAT/` after GSP. Without this the oriented cross sections came out as garbage (negative C_{ext}). This corrects a premise stated in the planning note, that the converged T-matrix simply remains available after the solve.

7.3 Geometric-optics regime, $x > 50$

Here the T-matrix does not converge and an asymptotic model is used. Extinction is the extinction paradox, $C_{\text{ext}} \rightarrow 2A_{\text{proj}}$, twice the geometric shadow. For an oblate spheroid of equatorial semi-axis a_s and symmetry semi-axis c ($a_s/c = b/a$), the shadow is πa_s^2 when $\mathbf{k}$ is along the symmetry axis ($\text{jori} = 1$) and $\pi a_s c$ when it is across ($\text{jori} = 2, 3$), giving $Q_{\text{ext}}(1) = 2(b/a)^{2/3}$ and $Q_{\text{ext}}(2) = Q_{\text{ext}}(3) = 2(b/a)^{-1/3}$. The shadow depends on the direction of $\mathbf{k}$ relative to the axis but not on the incident polarization, so extinction carries no polarization at this order.

The polarization in this regime is carried by absorption. In the opaque limit, where a refracted ray is fully absorbed — valid when $\text{Im}(m)x \gg 1$ — the absorptivity of an illuminated surface element is one minus its Fresnel power reflectance, and that reflectance is polarization-dependent ($R_s \neq R_p$ away from normal incidence). Integrating over the illuminated half of the spheroid,

$$C_{\text{abs},p} = \int_{\text{illum}} \left[1 - (|\hat{e} \cdot \hat{s}|^2 R_s(\theta_i) + |\hat{e} \cdot \hat{p}|^2 R_p(\theta_i)) \right] dA_{\text{proj}},$$

where θ_i is the local angle of incidence and $\hat{s}, \hat{p}$ the local senkrecht and parallel directions. The two grain-frame polarizations ($\hat{e} = \hat{a}$ versus $\hat{e} \perp \hat{a}$) project differently onto the local $\hat{s}, \hat{p}$ basis as they sweep the curved surface, and that is the whole source of the dichroism. A purely geometric ray treatment, with no Fresnel coefficients, would give no polarization at all; the surface reflectance is what makes the regime dichroic, and it is needed to match the release, which carries a real absorption dichroism there — about 3% of the mean, and *growing* with size parameter, the wave-optics signature a ray limit cannot reproduce. This is a deliberate approximation, documented at the code site with its opaque-limit domain of validity; it would matter for a size distribution with large-grain weight, which the astrodust distribution does not have.

7.4 Dispatch, generation, and drop-in

`oriented_cross_sections` selects the regime by x and returns the oriented ($Q_{\text{ext}}, Q_{\text{abs}}, Q_{\text{sca}}$) for one node, with the same branch boundaries and fallbacks as the random-orientation driver. `run_q_jori.x` loops it over the DH21 grid and writes the table in the exact stream the release reader expects (§2.), so the result is a drop-in: `sed_init` and `build_astrodust` already take an optional `qpol_path` that defaults to the release file, and pointing it at the regenerated table switches the polarized optics to SEDust’s own with no code change. A full sweep is dominated by the T-matrix nodes (of order 16 hours on one core), so the driver has a wavelength-window range mode for process-level parallelism and a merge mode that reassembles the windows — a window is not a contiguous slice of the stream, because the wavelength index runs innermost.

7.5 How far it has been checked

Each regime was compared node by node against the release table (Table 9). Across the Rayleigh and T-matrix regimes — every wavelength and radius that carries astrodust weight — the reconstruction matches the release to a median of a few parts in 10^4 , in both the mean and the polarized cross sections. The geometric-optics region agrees only coarsely, as an asymptotic model should: the mean absorption to about ten percent and the dichroism to about three-quarters of the release value, both with the correct sign and size-parameter trend. It joins the T-matrix continuously across $x = 50$ in sign and ordering, and it carries negligible weight in the size integral.

End to end, feeding the regenerated table through `calc_polext` reproduces the released polarized extinction to a median of 0.06% over $\lambda \geq 0.11 \mu\text{m}$, against 0.03% for the release table itself; the two agree with each other to a median of 0.02% at the level of the observable. So the few-parts-in- 10^4 reconstruction error at the level of a single cross section survives the size integral as a few parts in 10^4 in the polarized extinction, computed entirely from the astrodust dielectric function with no recourse to the release optics.

8. Aligned-grain scattering matrix and the extinction matrix

Section 7. regenerated the orientation-resolved *cross sections*. This section adds the angle-resolved object they left out: the fixed-orientation Mueller matrix $Z(\theta_i; \theta_s, \phi)$ of the aligned astrodust spheroid, and the 4×4 extinction matrix $K(\theta_i)$ that goes with it. Between them they complete the material side of the Peest-formalism contract — SEDust returns

every quantity in matrix form; the RT code (MoCafe, whose consumption side is not yet built) does the rotations, sampling, peel-off, and $\exp(-K\tau)$ transfer. The design is set out in docs/tmatrix_aligned_scattering_plan.md; this section records how each piece was built and why.

8.1 The engine

tmatrix/driver/scattering_matrix_oriented.f90 returns the fixed-orientation Mueller matrix Z of the DH21 oblate spheroid ($b/a = 1.4$) for one incidence direction. Two regimes, one output convention.

T-matrix regime, one AMPL call. With the T-matrix already converged and left in COMMON /TMAT/ by TMD_ONE_SCATMAT — the same restore of §7. that made the oriented cross sections possible — a single call to Mishchenko’s AMPL evaluates the 2×2 complex amplitude matrix $\mathbf{S} = [[S_{vv}, S_{vh}], [S_{hv}, S_{hh}]]$ for the chosen incidence and scattering angles. The 16 phase-matrix elements then follow from the standard bilinears in $\mathbf{S}$, copied *verbatim* from Mishchenko’s src/ampld.lp.f (its Z11...Z44, Eqs. (13)–(29) of Appl. Opt. **39**, 1026, 2000) so the phase matrix can be checked line by line against the source. Copying rather than rederiving is deliberate: the bilinears carry sign and conjugation conventions that are easy to transcribe wrongly, and the whole point of anchor C below is that this transcription is exact.

Rayleigh regime, analytic dipole. Below $x = 0.1$ the same Z is the electric-dipole matrix, $S_{pq} = k^2 (\hat{e}_p^{\text{sca}} \cdot \alpha \cdot \hat{e}_q^{\text{inc}})$ with the spheroid polarizability $\alpha = \text{diag}(\alpha_b, \alpha_b, \alpha_a)$, then the same bilinears. The polarizability comes from spheroid_dipole_polarizability, a routine extracted from rayleigh_limit (§7.) without changing an operation, so the dipole cross sections that §7. already validated and the dipole matrix here rest on one implementation of α_a, α_b .

The Stokes basis, stated once. Both regimes use Mishchenko’s meridional basis of each propagation direction *in the grain frame*, whose z axis is the alignment axis: $\hat{e}_1 = \hat{\theta}$ (call it v), $\hat{e}_2 = \hat{\phi}$ (h), with $\mathbf{Q} = I_v - I_h$. This is exactly the basis AMPL rotates its amplitudes into, so the analytic dipole and the T-matrix matrix share one convention with no adaptor between them. At $\theta_i = 90^\circ$ the v -polarization is jori = 2 ($\mathbf{E} \parallel \hat{a}$) and h is jori = 3 ($\mathbf{E} \perp \hat{a}$), which is what ties the extinction matrix back to the jori cross sections of §2..

8.2 Symmetries, verified not assumed

Three exact symmetries reduce the stored ranges and, more usefully, are numerical checks on the engine (anchor E). Each was confirmed on a grid of $(\theta_i, \theta_s, \phi)$ before being relied on.

- **Azimuth mirror.** The plane through the axis and the incidence direction is a symmetry plane, so $\phi \rightarrow 360^\circ - \phi$ leaves Z unchanged except that the two off-diagonal 2×2 blocks (elements 13, 14, 23, 24, 31, 32, 41, 42) flip sign. Store $\phi \in [0, 180^\circ]$; residual 3.5×10^{-7} .
- **Equatorial mapping.** The oblate spheroid is symmetric under $z \rightarrow -z$, so $\theta_i \rightarrow 180^\circ - \theta_i$ maps onto the stored

range with $\theta_s \rightarrow 180^\circ - \theta_s$, ϕ unchanged, and the same off-diagonal-block sign flip,

$$Z(180^\circ - \theta_i; 180^\circ - \theta_s, \phi) = S \odot Z(\theta_i; \theta_s, \phi), \quad S = \begin{pmatrix} + & + & - & - \\ + & + & - & - \\ - & - & + & + \\ - & - & + & + \end{pmatrix},$$

the diagonal blocks keeping their sign. Store $\theta_i \in [0, 90^\circ]$; residual 8.8×10^{-16} . Physically this also encodes that the alignment axis is headless.

- **Axisymmetry at $\theta_i = 0$.** There Z must be independent of ϕ up to a frame rotation, $Z(0; \theta_s, \phi) = Z(0; \theta_s, 0)R(\phi)$, the $\phi = 0$ matrix being six-element block-diagonal. This is a check, not a storage saving.

The library reconstructs the unstored octants from these relations, applying the ± 1 sign once to the interpolated matrix rather than to eight interpolation corners (§8.5); because each reflection contributes a constant that is the same for every corner of a cell, it factors out of the linear interpolation exactly.

8.3 The generator: one code path

The shared physics lives in `tmatrix/driver/aligned_population_optics.f90`, and both the production driver `run_scattermat_aligned.f90` and the verification program `compare_scattermat_aligned.f90` call it. This is on purpose: the numbers an anchor checks are produced by the exact code the production file is written from, so a passing anchor certifies the shipped path and not a parallel one.

One solve, three products. For each (band, size) in the T-matrix regime a single TMD_ONE_SCATMAT solve serves everything: its GSP expansion coefficients feed the random-orientation accumulation (the F block below), the `/TMAT/` it leaves valid feeds the AMPL sweep over the Z nodes, and its forward amplitudes give the extinction-matrix elements. No quantity is recomputed.

Regime split and the $x > 50$ guard. The size-parameter regimes match `run_scattermat.f90`: $x < 0.1$ analytic dipole, $0.1 \leq x \leq 50$ T-matrix. Above $x = 50$ no converged oriented matrix exists, so the bin is *omitted from the aligned products* and instead enters F_{tot} as its random-orientation geometric-optics matrix, i.e. scattered as unaligned. Its C_{sca} is tallied, and the driver stops hard if the omitted fraction of the scattering weight exceeds 10^{-6} . It never does for the shipped bands: the measured omitted fraction is at most 3.1×10^{-15} (at $0.36 \mu\text{m}$) and exactly zero at 0.64 and $0.79 \mu\text{m}$. The omission is thus provably negligible rather than assumed so.

An OpenMP lesson: -frecursive for the amplitude routine. The only parallel region is the loop over Z nodes inside one size. AMPL and VIGAMPL only *read* COMMON `/TMAT/` and otherwise write local automatic arrays, so distinct nodes are independent and each grid slot is written once — which is why a 1-thread and an N -thread run are bitwise identical. The subtlety is in the build: `ampl_oriented.f` is compiled `-frecursive` so its large local arrays become thread-private on the stack, while the serial T-matrix core (`tmd_one.f`, `lpd.f`) keeps its static locals under the base flags. Compiling the amplitude routine with static locals would have shared those arrays across threads and silently

corrupted the matrices; the Makefile therefore compiles that one file with the OpenMP-safe flags and the core without them. This is the implementation lesson worth carrying forward: thread-safety of a COMMON-based Mishchenko routine is decided by the compile flags of each file, not by a source edit.

8.4 Products and the η contract

Per band the file (output/scatmat_aligned_astrodust_P0.20_Fe0.00_1.400.dat) carries three blocks, all per H.

- **K block**, on the θ_i grid: $C_{\text{ext,al}}$, the dichroic $C_{\text{pol,al}}$, and the birefringent $C_{\text{bir,al}}$ from the forward amplitudes, plus $C_{\text{sca,al}}$ as the grid closure $\int Z_{11} d\Omega$. Because these come from the forward amplitude at each θ_i , the extinction matrix is *exact at every propagation angle*. This replaces the $\sin^2 \psi$ interpolation between the $\psi = 0$ and 90° values that the three-orientation jori table forces, and reduces to that law analytically in the Rayleigh limit (anchor G).
- **F block**: F_{tot} (every grain, identical to run_scatmat.f90) and F_{ref} (f_{align} -weighted), the six-element random-orientation matrices, α_1 -normalized so $\frac{1}{2} \int F_{11} d \cos \Theta = 1$; the header gives $C_{\text{sca,tot}}$, $C_{\text{sca,ref}}$, $C_{\text{ext,tot}}$, $C_{\text{ext,ref}}$ to restore absolute units.
- **Z block**: $Z_{\text{al}} = \int da n(a) f(a) Z(a)$, 16 elements, $\mu\text{m}^2 \text{sr}^{-1}$ per H, on $\theta_i = 0(5)90^\circ (19) \times \theta_s = 0(1)180^\circ (181) \times \phi = 0(5)180^\circ (37)$.

The η contract. The file header documents, and the library enforces, the exact rule for varying the alignment degree cell to cell by a single scalar η (the local alignment scale, $f_{\text{cell}} = \eta f_{\text{ref}}$). The aligned matrix scales linearly, $Z_{\text{al,cell}} = \eta Z_{\text{al}}$; the extinction matrix likewise, $K_{\text{al,cell}} = \eta K_{\text{al}}$; the unaligned remainder scattering matrix is $F_{\text{unal}} = F_{\text{tot}} - \eta F_{\text{ref}}$ in absolute units (after restoring $C_{\text{sca,tot}}$ and $C_{\text{sca,ref}}$); and the unaligned population adds an isotropic $C_{\text{ext,tot}} - \eta C_{\text{ext,ref}}$ to the extinction (its polarized and birefringent parts vanish). All of this is exact by the linearity of the size integral in f_{align} , so three stored integrals give the scattering optics of any η with no re-integration.

8.5 The library: two strictly separated layers

sed/src/scatmat_aligned.f90 (module scatmat_aligned_mod) serves the table to a polarized RT host with the same lifecycle as the rest of the library: initialize once, query along the path.

Initialization (serial, once). `load_scatmat_aligned(path, status)` parses the table into module storage (about 81.5MB resident for the production grid) and `free_scatmat_aligned` releases it. It is wired into `sed_init` / `build_astrodust` through an optional `scatmat_path` argument — exactly the `qpol_path` pattern of §7., except that a `scatmat_path` that fails to load is an error (`status = 3`), because a host that asked for the table depends on it, whereas a missing polarized-optics table only disables polarized emission. This layer is not thread-safe and is called from serial code.

Path queries (pure reads, OpenMP-safe). Five routines write only their own outputs and read the loaded arrays, so they are safe to call concurrently from photon threads:

- `scatmat_band(lambda_um, iband, exact)` — nearest stored band, so the hot path takes an integer and no wavelength search runs per event;
- `extinction_matrix_aligned(iband, theta_i, eta, kmat)` — the 4×4 matrix with diagonal $\eta C_{\text{ext,al}}$, $K(1,2) = K(2,1) = \eta C_{\text{pol,al}}$, $K(3,4) = \eta C_{\text{bir,al}}$, $K(4,3) = -\eta C_{\text{bir,al}}$, θ_i folded into $[0, 90^\circ]$ because K is even under the equatorial reflection;
- `mueller_matrix_aligned(iband, theta_i, theta_s, phi, z)` — trilinear interpolation plus the symmetry reconstruction of §8.2, at the reference alignment ($\eta = 1$; the caller scales);
- `mueller_matrix_random(iband, big_theta, f_tot, f_ref)` — the two six-element random-orientation matrices;
- `scattering_cross_sections(iband, theta_i, eta, csca_aligned, csca_unaligned)` with $\text{csca_unaligned} = C_{\text{sca,tot}} - \eta C_{\text{sca,ref}}$.

The loaded grids and arrays are also exposed `public`, `protected`, so a host that prefers to inline its own lookups can take them once at startup and never call back. The whole cell dependence enters through just two scalars the host already holds: η , and $\theta_i = \arccos(\hat{k} \cdot \hat{B})$ from one dot product.

The alignment guard (status 4). The K and Z arrays were integrated over size once, under the profile recorded in the header; the sanctioned runtime variation is η , not a change of profile. So when a loaded scattering table is present and the host calls `dust_set_alignment` or `dust_set_alignment_profile` with a profile that differs from the recorded one, the setter returns a *new* status code 4. It is non-fatal: the f_{align} update still completes (the emission and $C_{\text{pol,ext}}$ channels re-integrate live and honor the new profile), and the code only signals that the loaded scattering matrices no longer correspond to the model's alignment. A different profile requires regenerating the table (`run_scatmat_aligned.x profile=FILE`), which takes minutes. A tabulated profile never matches the recorded analytic one and so always raises the flag.

8.6 Birefringence through dust_extinction

`dust_extinction` gained an optional `Cbir_ext` output, the f_{align} -weighted size integral of the jori table's 4th-block birefringence,

$$C_{\text{bir,ext}}(\lambda) = \sum_a dn_{\text{Ad}}(a) C_{\text{bir,ext}}(\lambda, a) f_{\text{align}}(a),$$

mirroring `Cpol_ext` exactly. It is zero when the loaded table carries no 4th block (as the release table does not), so the default path is unchanged, and it is the same birefringence the aligned K block carries at $\theta_i = 90^\circ$, reached here without loading the aligned scattering table. This closes the gap noted in the previous release, that `dust_extinction` did not yet return the birefringence.

8.7 Verification

Every anchor of the plan passes (Table 10). The engine anchors (A–G) run at single sizes through `compare_scattermat_aligned.x`; the library checks run on the production table.

Cost of the production run. The five UBVRI bands (0.36, 0.44, 0.55, 0.64, 0.79 μm) take about 8 minutes at 32 threads (about 26 minutes at 8); the file is 138 MB, 38 MB gzipped. The existing regressions are unchanged: `calc_pol_ext` reproduces the release to a median of 0.0296% as before, and every `sed`, `mc` and `tmatrix` target builds clean.

9. Limits and extension paths

These are limits of the published optics, not of the code.

Birefringence is now computed; the release table has none. The release table stores Q_{ext} , Q_{abs} and Q_{sca} only, no phase retardation, so from it $C_{\text{circ}} = 0$. The first-principles route of §7. already evaluates the fixed-orientation amplitude matrix (Mishchenko’s AMPL, in `tmatrix/src/ampl_oriented.f`) and used only its imaginary forward part, for extinction. Keeping the real part as well costs nothing — it comes from the same amplitude call — and gives the birefringence, the phase retardation that couples U and V . It is stored as an optional 4th block of the regenerated table, $Q_{\text{re}}(\text{jori}) = (4\pi/k) \text{Re}[S_{\text{fwd}}]/(\pi a^2)$, the real twin of Q_{ext} , from which $C_{\text{bir}} = \frac{1}{2}(Q_{\text{re},3} - Q_{\text{re},2})\pi a^2$. The reader exposes it as `qbir_ext` with a `has_bir` flag that stays `.false.` for an older 3-block table, so nothing about the default path changes. No astrodust reference for the birefringence exists, so it is certified internally by Kramers-Kronig: Re and Im of the forward-amplitude difference are a Hilbert pair, and the directly computed birefringence agrees with the Hilbert transform of the already-validated dichroism to a median $\sim 0.1\%$ over the interior of the grid (the transform itself self-tests to 2×10^{-5} on a Lorentz oscillator; the residual is the finite-range truncation). What remains is to return it through `dust_extinction` and to consume it in the $U \leftrightarrow V$ transfer term, both on the host side.

The aligned scattering matrix is now computed. The release gives the integrated Q_{sca} with no angular information, so scattering by an aligned spheroid could not be modeled from it. It is now modeled from first principles: the fixed-orientation Mueller matrix $Z(\theta_i; \theta_s, \phi)$ of the aligned astrodust spheroid is built from the same fixed-orientation amplitude (`ampl_oriented.f`) used above for extinction, size-integrated over the astrodust distribution with the alignment weight, and paired with the 4×4 extinction matrix $K(\theta_i)$ from the forward amplitudes. It is written for five optical bands by `run_scattermat_aligned.x` and served to a polarized RT host through a two-layer library API. §8. is the full account.

The random-orientation scattering matrix does exist, in five optical bands. This is a different object from the previous paragraph and should not be confused with it. Dropping the alignment, a randomly oriented spheroid with a plane of symmetry has six independent scattering-matrix elements rather than sixteen, and those *are* computed in the tree: `tmatrix/driver/run_scattermat.f90` restores Mishchenko’s MATR expansion, accumulates the single-size matrices over the HD23 size distribution with $C_{\text{sca}} dn/n_{\text{H}}$ weighting, and writes

tmatrix/output/scatmat_astrodust_P0.20_Fe0.00_1.400.dat: one block per wavelength, 181 rows at $\theta = 0, 1, \dots, 180^\circ$, with columns $F_{11}, F_{22}, F_{33}, F_{44}, F_{12}, F_{34}$. It answers what an unaligned grain population scatters; it does not answer the aligned question above.

The stored wavelengths are 0.36, 0.44, 0.55, 0.64 and $0.79\mu\text{m}$, roughly *UBVRI*. That set was chosen, and the full sweep deliberately not run, because the planned use is the *V* band first with one or two further optical bands possible, and neither the far infrared nor the ultraviolet is in scope. The whole 1129-point grid costs about 1.7 hours and would be spent almost entirely on wavelengths nothing is waiting for. Extending is cheap and needs no code change: `./run_scatmat.x all` does the full grid, or a list of wavelengths does a subset, at about 5.5s per wavelength.

Far infrared needs neither. In that regime the albedo is essentially zero, so the transfer requires only the extinction matrix and the emission vector, both fully determined by what is exported. The HAWC+ $154\mu\text{m}$ comparison lies inside this regime, so the far-infrared milestone is reachable with the present data and should be attempted before any scattering work.

Astrodust only. DL07 and Zubko have no polarized optics. `build_dl07` releases any polarized arrays left by a previous astrodust initialization, so `lamI_pol` returns zeros rather than stale values.

9.1 Cost of build_Cpol

Loading the orientation-resolved table is the most expensive part of model construction (Table 11).

The orientation-resolved arrays `qext_j`, `qabs_j` and `qsca_j` are deallocated as soon as the combinations of Eqs. (1)–(2) have been formed. They are intermediate quantities: nothing downstream of `build_Cpol` reads them, and every later step uses the five derived arrays only. Releasing them removes 13.7 MB, about two-thirds of what the table would otherwise hold, and leaves 7.6 MB resident for the rest of the run. The three arrays are private to `q_table_jori`, so no consumer can depend on their surviving the build.

Measured on `main_astrodust.x`, the maximum resident set size falls from 51232 kB to 40088 kB, a reduction of about 10.9 MB. The peak does not fall by the full 13.7 MB because it is a high-water mark: once the orientation-resolved arrays are gone, the later SED solution stages set the peak instead. The steady-state occupancy just after `sed_init` does drop by the full 13.7 MB.

10. Reproducing the numbers

From `SEDust/sed/`:

```
make calc_polext.x # standalone polarized-extinction check
./calc_polext.x # -> output/polarized_extinction_ours.dat
                  # lambda, ours, reference, ratio

make calc_kext_astrodust.x # full-pipeline extinction table
./calc_kext_astrodust.x # -> ../data/kext_astrodust_MW.dat, 8 columns
```



```
make libsedust.a # library, for the emission path
```

`calc_polext.x` prints the median and maximum deviation against the release directly, restricted to $\lambda \geq 0.11 \mu\text{m}$. Column 8 of the extinction table should reproduce it to the ASCII precision quoted in Table 7; if the two disagree by more than that, the $\log a$ interpolation in `build_Cpol` or the size grid it interpolates onto is the place to look, since that is the only step separating the two routes.

For the emission fractions of Table 8, build a model with `build_astrodust`, evaluate `J_Mathis` at $U = 1.585$, and call `dust_emission` with `lamI_pol` requested; the example in §5.4 of the user manual is the complete recipe.

11. References

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File / entity	Responsibility
sed/src/q_table_jori.f90	The reader. Decompresses, parses the three quantity blocks, validates (monotone axes, finite values, no trailing data), forms Eqs. (1)–(2), and supplies $f_{\text{align_hd23}}(a)$, Eq. (4). Publishes $q_{\text{pol_ext}}$, $q_{\text{pol_abs}}$ and $q_{\text{ran_*}}$; the orientation-resolved $q_{\text{ext_j}}/q_{\text{abs_j}}/q_{\text{sca_j}}$ are private and are deallocated once the combinations have been formed.
sed_astrodust.f90: build_Cpol	Interpolates Q_{pol} from the 169-node table axis onto the size-distribution grid in $\log a$, exactly as C_{abs} and C_{sca} are, multiplies by πa_{eff}^2 , and fills $f_{\text{align_ad}}$. Checks that the polarized and Q wavelength grids agree node for node.
sed_astrodust.f90: Cpol, Cpol_ext, falign_ad	Module-level arrays, (NLAM, NA) and (NA). Cpol is the <i>absorption</i> one and drives polarized emission; Cpol_ext is its extinction twin and drives dichroic extinction.
dust_model_mod.f90: grain_pop_t	Carries $C_{\text{pol}}(:, :)$ and $f_{\text{align}}(:)$. Both unallocated is how the solver recognizes an unpolarized population. Also carries the extinction-side $C_{\text{pol_ext}}(:, :)$ and $g_{\text{sca}}(:, :)$, read only by dust_extinction.
sed_astrodust.f90: set_pop	Optional Cpol_in, falign_in. Both must be supplied together; supplying neither leaves the population unpolarized. The extinction-side C_sca_in, Cpol_ext_in and g_sca_in are optional and independent; a population that omits them contributes zero to those terms of the size integral.
sed_astrodust.f90: sed_grain_loop	Optional Cpol_pop, falign_pop, Jpol, again all three or none (do_pol). Accumulates the polarized channel alongside Jout in every solver branch (§5.1).
sed_astrodust.f90: dust_emission	Optional lamI_pol, last argument. Sums Jpol over the populations that carry polarized optics and applies the same unit convention as lamI_total.
sed_astrodust.f90: dust_extinction	The extinction accessor. Integrates C_{abs} , C_{sca} , the scattering-weighted $\langle \cos \theta \rangle$ and $C_{\text{pol_ext}} f_{\text{align}}$ over the size distribution of every population, and returns them per H atom on $m\lambda_{\text{m}}$. These are the same numbers column 8 of the table carries, without the file (§5.7).
sed/src/calc_polext.f90	Standalone check: computes the polarized extinction directly from $q_{\text{pol_ext}}$ and the release size distribution, and compares against the published polarized_extinction.dat. Independent of sed_init.
sed/src/calc_kext_astrodust.f90	Writes column 8, $C_{\text{pol_ext}}/H$, of data/kext_astrodust_MW.dat as $\sum_a dn_{\text{Ad}}(a) C_{\text{pol_ext}}(\lambda, a) f_{\text{align}}(a)$.

Table 4: Responsibility of each file touched by the polarization update.

branch	context	accumulates into	pattern
'draine'	equilibrium grain	Jpol_local	thread-local
'draine'	stochastic, sum over $P(T)$	Jpol_local	thread-local
'heuristic'	equilibrium grain	Jpol	direct
'heuristic'	stochastic, sum over $P(T)$	Jpol	direct
'qm'	QM solve, rescaled	Jpol_local	thread-local
'qm'	GD fallback after QM failure	Jpol_local	thread-local
'qm'	equilibrium grain	Jpol_local	thread-local
'equil'	every grain	Jpol	direct

Table 5: Where the polarized emission is accumulated. The two boldface rows are the ones a search for the Jout_local pattern does not find.

band	median	maximum
far infrared ($\lambda > 30 \mu\text{m}$)	0.022%	3.46%
restricted to $a \leq 0.5 \mu\text{m}$		1.98%
mid infrared	0.033%	
near infrared	0.038%	
optical	0.080%	
ultraviolet	0.21%	9.53%

Table 6: Disagreement between the trace average and the exact orientation average of Q_{abs} .

Check	Result
Column 8 vs. released polarized_extinction.dat	median 0.0296%, max 0.9422%, 992 points
Column 8 vs. calc_polext.x	agree to 7×10^{-8} relative
Columns 1–7 of kext_astrodust_MW.dat	byte-identical to the pre-change file
astrodust_irem_ours_S1/S2/PAH.dat	byte-identical
dl07_sed_ours_mw31_60.dat	byte-identical
Build	zero warnings

Table 7: Verification of the polarized extinction path and regression of everything else.

λ	fraction
$12\,\mu\text{m}$	0.00%
$100\,\mu\text{m}$	13.67%
$154\,\mu\text{m}$	17.15%
$350\,\mu\text{m}$	18.79%
$850\,\mu\text{m}$	19.15%
median over $300\text{--}3000\,\mu\text{m}$	19.18%

Table 8: Intrinsic polarization fraction of the emitted spectrum, before any geometric factor.

Regime	x	Quantity	median ours/HD23 – 1
Rayleigh	< 0.1	$Q_{\text{ran}}(\text{abs})$	2.3×10^{-4}
		$Q_{\text{pol}}(\text{abs})$	3.7×10^{-4}
T-matrix	0.1–50	$Q_{\text{ran}}(\text{ext})$	2.1×10^{-4}
		$Q_{\text{pol}}(\text{ext})$	8.4×10^{-4}
		$Q_{\text{ran}}(\text{abs})$	1.9×10^{-4}
		$Q_{\text{pol}}(\text{abs})$	7.6×10^{-4}
GO	> 50	$Q_{\text{ran}}(\text{abs})$	9.3×10^{-2}
		$Q_{\text{pol}}(\text{abs})$	3.3×10^{-1}

Table 9: Node-by-node reconstruction against the HD23 release table. The geometric-optics dichroism is at ~ 0.75 of the release with matching sign and trend; its size-parameter region carries no astrodust weight.

Check	What it certifies	Result
<i>Engine (single size)</i>		
A closure $\int Z_{11} d\Omega$ vs $C_{\text{sca}}(\text{jori})$	phase-matrix normalization	2.7×10^{-7}
B optical theorem vs $C_{\text{ext}}(\text{jori})$	forward-amplitude geometry	2×10^{-14}
C orientation average vs GSP $F(\Theta)$	the bilinears, incl. F_{12}, F_{34} signs	1.4×10^{-6}
D dipole vs closed form	Rayleigh implementation	2×10^{-16}
dipole vs T-matrix at $x \sim 0.1$	regime continuity, $ \delta Z /Z_{11}$	0.5–1.1%
E ϕ mirror	azimuth symmetry	3.5×10^{-7}
equatorial mapping	$z \rightarrow -z$ symmetry	8.8×10^{-16}
G Rayleigh $K(\theta_i)$ vs $\sin^2$ law	extinction-matrix angle law (exact)	3.3×10^{-16}
T-matrix $K(\theta_i)$ vs $\sin^2$ law	at $x = 0.137$	6.3×10^{-4}
<i>Generator (size-integrated)</i>		
F K vs independent jori integrals	C_{ext}	1.6×10^{-6}
	C_{pol}	2.0×10^{-5}
	C_{bir}	1.3×10^{-7}
<i>Library (production table)</i>		
reader round-trip	parse vs re-emit	bitwise 0
symmetry reconstruction	unstored octants	bitwise 0
η algebra	linear scaling	residual 0
$K(90^\circ)$ vs jori vs dust_extinction	$C_{\text{pol}} / C_{\text{bir}}$	$1.6 \times 10^{-5} / 8.0 \times 10^{-5}$
rt_example closures	end-to-end consumption	0.997–1.001

Table 10: Verification of the aligned scattering matrix, its extinction matrix, and the library API. Engine and generator anchors are those of the plan’s §5; the library checks run on the shipped production table.

Item	Size
compressed file	4.1 MB
scratch file when expanded	17.2 MB
raw arrays qext_j, qabs_j, qsca_j	13.7 MB, transient
derived qpol_*, qran_*	7.6 MB, kept
resident after sed_init	7.6 MB
time	about 1 s of the 3.1 s model build

Table 11: Cost of loading the orientation-resolved table. The orientation-resolved arrays exist only between the read and the formation of the derived combinations.